

Elia Piccoli

PHD IN COMPUTER SCIENCE

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Education

PhD in Computer Science, UNIVERSITY OF PISA, ITALY

Nov. 2022 - Apr. 2026

- Title: *Knowledge in Robust and Lifelong Reinforcement Learning Agents*
- Supervisors: Prof. Davide Bacciu, Prof. Vincenzo Lomonaco
- Themes: *Reinforcement Learning, Lifelong Learning, Knowledge Representation, Compositionality, Safety*

ESSAI & ACAI Summer School, LJUBLJANA, SLOVENIA

Jul. 2023

- European Summer School on Artificial Intelligence & Advanced Course on Artificial Intelligence.
- Four advanced courses (40 hours): *AutoML: Accelerating Research on and Development of AI Applications; Learning to Act and Plan; Multi-Objective Reinforcement Learning; Game-Theoretic Approach to Planning and Synthesis.*

Master Degree in Computer Science, UNIVERSITY OF PISA, ITALY

Sep. 2020 - Oct. 2022

- Curriculum: *Artificial Intelligence*
- Thesis Title: *Introducing Unsupervised Skills in Continual Reinforcement Learning Agents*
- Final Degree: 110/110 Hons. (equiv. GPA: 4/4)

CyberChallengeIT hosted by CINI, UNIVERSITY OF VERONA, ITALY

Feb. 2020 - Jun. 2020

- The course was divided into two parts: (i) *theoretical* section analyzing all the various fields that are strictly correlated to cybersecurity (e.g. cryptography, binary exploitation, web hacking, etc.); (ii) *hands-on sessions* on how to approach and analyze different problems, how to spot possible exploits and take advantage of them to retrieve data or access to machines.
- Selection phase acceptance rate 15%.

Bachelor Degree in Computer Science, UNIVERSITY OF VERONA, ITALY

Sep. 2017 - Jul. 2020

- Thesis Title: *Deep Reinforcement Learning for Multi-Agent Navigation*
- Final Degree: 110/110 Hons. (equiv. GPA: 3.96/4)

Experience

Visiting Researcher, UNIVERSITY OF ALBERTA, CANADA

Mar. 2026 - PRESENT | Apr. - Sep. 2025

- Themes: Reinforcement Learning, Programmatic Policies, Program Synthesis with LLMs, Search Algorithms.
- Supervisor: Prof. Levi Lelis

Research Collaboration, APTUS.AI, ITALY

Aug. 2024 - Feb. 2025

- Worked with Large Language Models on document retrieval to improve Q&A pipelines for Italian statutory laws.
- The work contributed to the initial development of a project that later resulted in a scientific publication (EACL 2026).

Teaching Assistant, UNIVERSITY OF PISA, ITALY

Sep. - Dec. 2024 | Sep. - Dec. 2023

- Course: *Artificial Intelligence Fundamentals* (Master Degree in Computer Science, Curriculum AI)
- Hands-on lessons throughout the course over several topics such as: search algorithms, knowledge representation, reasoning & planning, reasoning under uncertainty.

Teaching Assistant, UNIVERSITY OF PISA, ITALY

Feb. 2023 - Jun. 2023

- Course: *Computer Architecture and Operating Systems* (Bachelor Degree in Computer Science)
- Hands-on lessons solving exercises in preparation for intercourse exams and reinforcement of learned concepts.

Autonomous Planning Engineer, E-TEAM SQUADRA CORSE, PISA

Oct. 2022 - Jun. 2024

- Division: *Driverless, AI & Software Development.*
- Worked on building the entire training pipeline using Reinforcement Learning to create an autonomous driving system capable of driving the car on a Formula SAE tracks. We also worked on the interface with the CARLA simulator, also using parallel environments, and deploying the whole training loop via Python and PyTorch.

University Internship, UNIVERSITY OF VERONA, ITALY

Nov. 2019 - Feb. 2020

- Created a customizable Unity game environment that generates random mazes and used it as learning environment to train agents via Python and Keras using ml-agents tool. The created environment included both single and multiple agents scenarios.
- In depth analysis and comparison of different state-of-the-art Reinforcement Learning algorithms using several environments.

Tutoring, HIGH SCHOOL AND UNIVERSITY STUDENTS

2016 - PRESENT

- Worked in total with more than 30 students helping them studying and understanding the topics. The subjects include Computer Science, Math and Physics.

Publications

- G. Aru*, **E. Piccoli***, F. Trotti*, A. Farinelli, D. Bacciu. *Uncertainty-Gated Safety for Out-of-Distribution Map-less Navigation*. In [Under Review](#). (*** Equal Contrib.**)
- **E. Piccoli**, M. Li, G. Carfi, V. Lomonaco, D. Bacciu. *Combining Pre-Trained Models for Enhanced Feature Representation in Reinforcement Learning*. In [IBRL workshop, RLC 2025](#) and [CoLLAs 2025](#).
- A. Capurso*, **E. Piccoli***, D. Bacciu. *FAST: Similarity-based Knowledge Transfer for Efficient Policy Learning*. In [IEEE CoG 2025](#). (*** Equal Contrib.**)
- J. Bell, L. Quarantiello, E.N. Coleman, L. Li, M. Li, M. Madeddu, **E. Piccoli**, V. Lomonaco. *The Future of Continual Learning in the Era of Foundation Models: Three Key Directions*. In [TCAI workshop, HHAI 2025](#).
- L. Quarantiello*, **E. Piccoli***, et. al. *A Compositional Paradigm for Foundation Models: Towards Smarter Robotic Agents*. In [I-RIM 3D 2025](#). (*** Equal Contrib.**)
- M. Li*, **E. Piccoli***, V. Lomonaco, D. Bacciu. *I Know How: Combining Prior Policies to Solve New Tasks*. In [IEEE CoG 2024](#). (*** Equal Contrib.**)
- L. Li*, **E. Piccoli***, A. Cossu, D. Bacciu, V. Lomonaco. *Calibration of Continual Learning Models*. In [CLVision workshop, CVPR 2024](#). (*** Equal Contrib.**)

Talks, Posters & Presentations

Oral Presentation, IEEE CoG 2025

Aug. 2025

- Presentation of the paper: *FAST: Similarity-based Knowledge Transfer for Efficient Policy Learning*.

Poster Presentation, IBRL WORKSHOP (RLC 2025) AND COLLAS 2025

Aug. 2025

- Poster presentation of the paper: *Combining Pre-Trained Models for Enhanced Feature Representation in Reinforcement Learning*.

Oral Presentation, IEEE CoG 2024

Aug. 2024

- Presentation of the paper: *I Know How: Combining Prior Policies to Solve New Tasks*.

Poster Presentation, CLVISION WORKSHOP (CVPR 2024)

Jun. 2024

- Poster presentation of the paper: *Calibration of Continual Learning Models*.

Talk at Continual Learning Course, UNIVERSITY OF PISA

May 2024

- Seminar for the course presenting the paper: *Calibration of Continual Learning Models*.

Talk at Continual Learning Course, UNIVERSITY OF PISA

May 2023

- Seminar for the course titled: *Introduction on Continual Reinforcement Learning*.

Student Supervision

Giacomo Aru, PH.D. IN ARTIFICIAL INTELLIGENCE, UNIVERSITY OF PISA

2025 - PRESENT

- Working on Reinforcement Learning, Multi-Agent Systems, Graph Neural Networks, Uncertainty Estimation.
- Co-supervised with *Davide Bacciu, Enrico Marchesini*.

Malio Li, PH.D. IN COMPUTER SCIENCE, UNIVERSITY OF PISA

2024 - PRESENT

- Working on Causal Reinforcement Learning, Causal World Models, Option Learning.
- Co-supervised with *Davide Bacciu, Riccardo Massidda*.

Giacomo Aru, MASTER DEGREE IN COMPUTER SCIENCE, CURRICULUM AI, UNIVERSITY OF PISA

2025

- Title: *Uncertainty-Aware Safe Reinforcement Learning using Control Barrier Functions*
- Co-supervised with *Davide Bacciu, Alessandro Farinelli, Francesco Trotti*. Collaboration with University of Verona.

Marco Petix, MASTER DEGREE IN COMPUTER SCIENCE, CURRICULUM AI, UNIVERSITY OF PISA

2025

- Title: *Behaviour Characterization via π -vectors in Multi-Task Reinforcement Learning*
- Co-supervised with *Davide Bacciu*

Alessandro Capurso, MASTER DEGREE IN COMPUTER SCIENCE, CURRICULUM AI, UNIVERSITY OF PISA

2024

- Title: *Similarity-Based Policy Transfer for Multi-Task Reinforcement Learning*
- Co-supervised with *Davide Bacciu*

Giacomo Carfi, MASTER DEGREE IN COMPUTER SCIENCE, CURRICULUM AI, UNIVERSITY OF PISA

2024

- Title: *Adaptively Combining Skill Embeddings for Reinforcement Learning Agents*
- Co-supervised with *Davide Bacciu*

Malio Li, MASTER DEGREE IN COMPUTER SCIENCE, CURRICULUM AI, UNIVERSITY OF PISA

2024

- Title: *Solving multiple tasks simply using prior policies in Lifelong Reinforcement Learning*
- Co-supervised with *Vincenzo Lomonaco*

Alessandro Guerriero, BACHELOR DEGREE IN COMPUTER SCIENCE, UNIVERSITY OF PISA

2023

- Title: *Search Algorithms as Skills for Reinforcement Learning agents.*
- Co-supervised with *Davide Bacciu*

Projects

Here are reported some of my personal projects, all of them - including the research ones - can be found on my [GitHub](#).

Minesweeper RL environment in C

2025

- Used [C](#) to implement the game logic, [Raylib](#) for visualization and [PufferLib](#) to train RL agents to play the game.

SmartPA

2021

- The project was developed in collaboration with *Compagnia Trasporti Toscana (CTT)*. The focus of the project was to improve the document archiving process in Public Administration exploiting Artificial Intelligence.
- We created from scratch a Transformer architecture, using [Python](#) and [PyTorch](#), to predict the archiving class for the documents. Using real data from the last two years, we were able to obtain models with more than 80% top3-accuracy.
- The collaboration and results were also featured on newspaper: [Article link](#)

Crossword Generation

2020

- The program is written in [Python](#) and uses [Flask](#) for a naive web interface. Given a set of words and their definition it builds a crossword placing all the words, its [SVG](#) representation and the match between crossword number - definition.

Defect Detection using Spectral Analysis

2020

- Used [Matlab](#) and [spectral analysis](#) to analyze and highlight texture defects on images.

Skills

Programming Languages	Python, C, C++, Java, MATLAB & Simulink, Bash, SQL
Libraries/Frameworks	PyTorch, Keras, TensorFlow, Numpy, Sklearn, Pandas, Huggingface Transformers
Softwares	Visual Studio, Git, Microsoft Office, PostgreSQL, Unity, Blender, Docker

Funding

2025	Research Grant (€8,200) , Project: <i>Development of compositional Reinforcement Learning models.</i>	<i>University of Pisa</i>
2022	Fully Funded PhD Scholarship , 3-year, funded by PNRR.	<i>University of Pisa</i>

Languages

Italian	Native speaker
English	Fluent First Certificate in English (B2 - 2016) English for Research Publication and Presentation (C1 - 2023)
Spanish	Elementary

Extra

PhD Student Representative, UNIVERSITY OF PISA, ITALY

Nov. 2023 - Oct. 2025

- Student representative on the *PhD Steering Committee in Computer Science*.

Returning Officer, COMUNE DI SAN GIOVANNI LUPATOTO, VERONA

2018 - 2025

- Management and supervision of voting operations, ballot counting and compilation of minutes.

Personal Data Treatment

I hereby authorize the processing of my personal data in accordance with the current data protection regulations, specifically the European General Data Protection Regulation (GDPR) 2016/679, the Italian Legislative Decree No. 196 of 30 June 2003, and any subsequent amendments and additions.